LangChain Learning Roadmap

2-Week Curriculum

10 Days — LangChain Core (v1.0)

Day 1 — Setup + Your First Agent

Docs Reference: Install, Quickstart — docs.langchain.com

Concepts: What LangChain is, LLM providers, create_agent()

Key Idea: Build a working agent in under 10 lines of code

🚀 Mini Project: "Hello Agent" — ask a question, get an answer from an LLM

Day 2 — Models & Messages

Docs Reference: Models, Messages — docs.langchain.com/oss/python/langchain/models

Concepts: Standardized model interface, provider switching, message formats

Key Idea: LangChain lets you swap OpenAI, Anthropic, Google seamlessly — no lock-in

🚀 Mini Project: "Model Switcher" — same prompt, compare outputs from 2 different LLM providers

Day 3 — Tools

Docs Reference: Tools — docs.langchain.com/oss/python/langchain/tools

Concepts: Custom tool functions, @tool decorator, name/description/signature

Key Idea: Tools extend the agent — APIs, databases, file readers, web browsers

🚀 Mini Project: "Weather Bot" — agent that calls a custom weather tool function.

Day 4 — Structured Output

Docs Reference: Structured Output — docs.langchain.com/oss/python/langchain/structured-output

Concepts: JSON responses, Pydantic models, typed structured data from LLMs

Key Idea: Get predictable, parseable output instead of free-form text

🚀 Mini Project: "Movie Recommender" — returns a structured list of movies with title, genre & reason

Day 5 — Short-term Memory

Docs Reference: Short-term Memory — docs.langchain.com/oss/python/langchain/short-term-memory

Concepts: Conversation history, stateful agents, multi-turn context

Key Idea: Agent remembers what was said earlier in the same conversation

🚀 Mini Project: "Chatbot that Remembers" — multi-turn conversation with persistent context

Day 6 — Retrieval (RAG Foundation)

Docs Reference: Retrieval — docs.langchain.com/oss/python/langchain/retrieval

Concepts: Document loaders, text splitting, embeddings, vector stores (FAISS/Chroma)

Key Idea: RecursiveCharacterTextSplitter chunks docs; embeddings enable similarity search

🚀 Mini Project: "Semantic Search over PDF" — load a PDF and search it by meaning, not keywords

Day 7 — RAG Agent

Docs Reference: RAG Agent Tutorial — docs.langchain.com/oss/python/learn

Concepts: RetrievalQA chain, combining retrieval + agent loop end-to-end

Key Idea: Ground LLM answers in real documents to reduce hallucinations

🚀 Mini Project: "Chat with your PDF" — ask any question about an uploaded document

Docs Reference: Context Engineering, Middleware — docs.langchain.com/oss/python/langchain

Day 8 — Context Engineering + Middleware

Concepts: What LLM sees per call vs. what persists in state; lifecycle hooks

Key Idea: Dynamically inject user profile, date, rules into every agent call

🚀 Mini Project: "Smart Prompt Injector" — middleware that adds context to every agent request

Day 9 — Multi-Agent Systems

Docs Reference: Multi-agent — docs.langchain.com/oss/python/langchain/multi-agent

Concepts: Subagent delegation, handoffs, routing queries to specialised agents

Key Idea: A main orchestrator agent delegates tasks to focused sub-agents

🚀 Mini Project: "Personal Assistant" — main agent delegates to Search Agent & Calculator Agent

Day 10 — LangGraph Introduction

Docs Reference: LangGraph Overview — docs.langchain.com/oss/python/langgraph/overview

Concepts: Graph-based workflows, nodes/edges, state, human-in-the-loop, durable execution

Key Idea: LangGraph gives fine-grained control for complex agentic pipelines

🚀 Mini Project: "Custom RAG Agent" — rebuild RAG with LangGraph, adding a human approval step

💼 LangChain Ecosystem Map (v1.0)

Layer	Tool	Purpose
Framework	LangChain	Build agents fast with prebuilt architecture
Orchestration	LangGraph	Complex, stateful, customizable workflows
Production	Deep Agents	Batteries-included agents for production
Observability	LangSmith	Debug, trace, and evaluate agent behaviour